

ACTIVITY 2 - Real-Life Scenario – Role-Play

In your group, perform a short role-play based on a real-life situation where a safety device prevents injury or reduces damage during an accident or high-impact event. Your scene should show how the safety device works and explains the physics behind its protective function.

Assigned Scenarios:

Group 1: A biker loses control and falls off the bike but is wearing a helmet.

Group 2: A person slips on a wet floor but avoids injury due to wearing elbow/knee pads.

Group 3: A car crashes into a wall, but airbags deploy and protect the driver.

Group 4: A gymnast falls during practice, but lands on a cushioned mat.

Group 5: A child jumps off a trampoline and hits the safety net instead of falling to the ground.

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Each group must include:

- A brief presentation of what happened and who was involved
- Identification of the safety device used
- Explanation of the physics concept involved, such as:

How the device increases the time of impact?

How this reduces the force experienced by the person?

What principle is applied (e.g., momentum, impulse, energy absorption, Newton's Laws)?